

4.4 Prevent “shoulder surfing”

In computer security, shoulder surfing refers to using direct observation techniques, such as looking over someone’s shoulder, to get information. Shoulder surfing is particularly effective in crowded places because it’s relatively easy to observe someone as they enter passwords on a computer, fill out a form or enter their PIN at an automated teller machine (ATM).

Shoulder surfing can also be done at a distance using binoculars or other vision-enhancing devices. Inexpensive, miniature closed circuit television cameras can be concealed in ceilings, walls or fixtures to observe data entry. To prevent shoulder surfing, it is advised to shield paperwork or the keypad from view by using one’s body or cupping one’s hand. When working on a laptop, ensure that your back is to a wall with no open sides close to you or to enter your passwords in a secluded location.

Recent automated teller machines now have a sophisticated display which discourages shoulder surfers. It grows darker beyond a certain viewing angle, and the only way to tell what is displayed on the screen is to stand directly in front of it.

Certain models of credit card readers have the keypad recessed, and employ a rubber shield that surrounds a significant part of the opening towards the keypad. This makes shoulder-surfing significantly harder, as seeing the keypad is limited to a much more direct angle than previous models. Taken further, some keypads alter the physical location of the keys after each key-press. For example the digit 1 may be the upper left on the first press, then moves to the bottom right for the second.

4.5 Encrypt your hard drive

Securing computer data through encryption software has become a clear necessity for many businesses and individuals carrying sensitive information on their laptops or USB flash drives.

Unfortunately, many people do not encrypt their data because it is not easy to implement in Windows and the learning curve is too high. However, thanks to open source initiatives, you can use a completely free program called TrueCrypt to encrypt an entire hard drive, a USB flash drive, or even create a virtual encrypted disk in a file that acts as a real disk.